or read data from database tables even when no other configuration within the class is written.

Active Record goes beyond simple persistence and includes lots of data modeling and data validation features. Active Record objects have *Associations* to link models together and *Validations* for validating the correctness of input data.

Active Record Terminology



- Object Relational Mapper (ORM) — Maps relational database tables and to Active Record model classes in Ruby on Rails
- Migrations — In Rails, Migrations are incremental modifications to the database structure.
- Associations — Expressing data model relationships with Active Record using methods like `has_many`, `belongs_to`
- Validations — Application level consistency checking, null/nil, format, model relationships

Now that some of the standard PostgreSQL, SQL and Rails terms are clear, let's look at conventions used in the coming chapters.

Conventions Used In This Book

Code snippets are used extensively for a variety of types of code like shell scripts, SQL, Ruby scripts, and more. Code snippets are available for download. The language is selected to help with coloring the keywords when viewing the text with color.

For CLI programs with options that support short and long option names, like `-v` or `--version` for version, the longer syntax is used.

Shell scripts start with `sh` in the filename. These are meant to be run in your Terminal program. On Mac OS, use either the Terminal.app terminal or a third party terminal like iTerm.[6]

For example, a shell script called `load_data.sh` would be run from your terminal by typing `sh load_data.sh`.

SQL keywords are capitalized and monospaced like `SELECT`, `WHERE`, and `LIMIT`. Portions of a query that are incomplete are referred as *clauses* or *fragments*, although full queries to run in a SQL client are preferred.

6. https://iterm2.com